

Apex Systems

Engineered for longevity, secured by design

Replacement of the Library Management System

Greenfield City Library Service operates a central library and eleven branch libraries serving approximately 184,000 registered members. The present library management system was commissioned in 2006, is no longer supported by its original vendor, and cannot support self-service borrowing, online renewals, or consolidated reporting across branches. The Service invites proposals for a replacement covering catalogue management, membership, circulation (borrowing, returns, renewals, reservations), inter-branch transfers, overdue fines, and public search.

Submitted to	Greenfield City Library Service
RFP reference	GCLS/RFP/2026/014
Submission date	2026-09-08
Proposed duration	9 months

FICTIONAL DOCUMENT -- prepared for an academic exercise. Greenfield City Library Service, the suppliers and all references named in this document are invented. No real organisation is described.

1. Executive Summary and Understanding of the Requirement

Apex Systems is pleased to respond to reference GCLS/RFP/2026/014 for the replacement of the Greenfield City Library Service management system. We understand that the incumbent system has been unsupported since 2019, that eleven branches currently reconcile circulation data by overnight batch file, and that members cannot renew a loan without attending a branch in person. Our reading of the requirement is that the Service is not buying a like-for-like replacement but a platform expected to remain in service for fifteen years.

We have therefore proposed an architecture that is deliberately conservative in its operational choices and modern in its technical ones. The result carries a higher capital cost than a minimal replacement, and we make no attempt to disguise that. We believe the difference is justified by the migration guarantee, the security posture, and the absence of a forced re-platforming in year five.

- Single consolidated catalogue across all twelve sites, no overnight batch reconciliation
- Self-service borrowing and returns at RFID kiosks, with staff override
- Public catalogue search, online renewals and reservations, on web and mobile
- Full audit trail on every member and item transaction, retained for seven years

2. Proposed Solution and Technical Approach

The solution is a service-oriented application deployed on a Kubernetes cluster in two geographically separated regions, with PostgreSQL 16 in a primary-replica configuration and automated failover. Circulation, catalogue, membership and reporting are separate services communicating over an internal API gateway, so a fault in reporting cannot take circulation offline. All services are stateless; session state is held in a replicated Redis cluster.

Every external interface is exposed as a documented REST API with OpenAPI 3.1 specifications supplied to the Service at handover. Integration with the existing 3M RFID gates and self-service kiosks uses SIP2 with a compatibility shim we have deployed at four comparable library authorities. Payments for overdue fines are handled through a PCI DSS compliant gateway; no cardholder data touches the library system. Staff authentication federates to the council Active Directory via SAML 2.0 single sign-on.

The platform is sized for a peak of 1,400 concurrent public catalogue sessions and 320 concurrent staff sessions, which is four times the Service's measured 2025 peak. Load testing evidence against these figures will be supplied before user acceptance testing begins. The system is designed to a recovery time objective of four hours and a recovery point objective of fifteen minutes.

- Kubernetes across two regions with automated failover; RTO 4 hours, RPO 15 minutes
- PostgreSQL 16 primary-replica; point-in-time recovery to any second in the last 35 days
- SIP2 and REST integration with existing RFID gates, kiosks and payment gateway
- SAML 2.0 single sign-on against council Active Directory
- OpenAPI 3.1 specifications and full source documentation delivered at handover

3. Implementation Plan, Team and Milestones

We propose a nine month programme. This is longer than some respondents will offer. The additional time is concentrated in migration and parallel running: eighteen years of catalogue and member data cannot be moved safely in a single weekend cutover, and we will not propose one.

Phase	Weeks	Key deliverable	Lead role
1. Discovery and design	1-6	Solution design document, signed off	Solution Architect
2. Build and configure	7-20	Configured platform in test environment	Technical Lead
3. Data migration trials	14-26	Three full migration rehearsals with reconciliation reports	Data Migration Lead
4. Integration and testing	21-30	RFID, payments and SSO integrated and tested	Integration Engineer
5. User acceptance testing	31-34	UAT sign-off by Service, defect log cleared	Test Manager
6. Parallel running	35-38	Four weeks live alongside incumbent system	Project Manager
7. Cutover and hypercare	39-40	Go-live, 4 weeks on-site hypercare	Project Manager

The core team is eight people: a Project Manager (0.6 FTE), Solution Architect (0.5 FTE), Technical Lead (1.0 FTE), two Developers (2.0 FTE), Data Migration Lead (0.8 FTE), Integration Engineer (0.5 FTE) and Test Manager (0.6 FTE). Named CVs are provided in Annex A. We commit contractually to the named Solution Architect and Data Migration Lead remaining on the engagement to go-live.

4. Commercial Proposal

Pricing is fixed-price for the implementation, with recurring charges held flat for the initial three year term and capped at CPI thereafter.

One-off implementation charges	Amount
Discovery, solution design and architecture	\$48,000
Platform licence, perpetual, 12 sites	\$145,000
Configuration and development	\$96,000
Data migration, 18 years of catalogue and member records	\$62,000
Integration: RFID, payment gateway, council SSO	\$28,000
Training and go-live support	\$16,000
Total one-off charges	\$395,000

Recurring annual charges	Amount
Managed hosting and infrastructure	\$34,000
Support and maintenance, 24x7, P1 response 1 hour	\$38,000
Security monitoring and annual penetration test	\$6,000
Total per year	\$78,000

Total cost of ownership	\$395,000 + 3 x \$78,000 = \$629,000
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Figures are quoted over a 3 year contract term and exclude sales tax.

Commercial assumptions:

- The Service provides a single point of contact empowered to sign off each phase
- Existing RFID gates and kiosks remain in place and are SIP2 capable
- Legacy data is made available in a readable export within four weeks of contract signature
- Prices exclude applicable sales tax and are valid for 90 days from submission
- Perpetual licence: the Service retains the right to use the platform if support lapses

5. Security, Compliance and Risk Controls

Apex Systems holds ISO 27001:2022 certification, most recently audited in March 2026, and completes an annual SOC 2 Type II examination. Certificates and the most recent SOC 2 report are available under NDA on request.

Member data is encrypted at rest using AES-256 and in transit using TLS 1.3. Access is governed by role-based access control with eleven predefined library roles; every privileged action writes an immutable audit record capturing actor, timestamp, and the before and after state of the record. Borrowing history is pseudonymised after the retention period configured by the Service, defaulting to two years, in recognition of the particular sensitivity of library borrowing records. Data is held exclusively in-country; no member data is processed outside the jurisdiction.

An independent CREST-accredited penetration test is performed before go-live and annually thereafter, with the executive summary shared with the Service and all high or critical findings remediated within 30 days. We maintain a documented incident response procedure with a 24 hour breach notification commitment to the Service.

Risk	Likelihood	Mitigation
Legacy data quality worse than expected	High	Three migration rehearsals; reconciliation report signed off before each
RFID kiosks not SIP2 compliant	Medium	Compatibility assessed in week 2 of discovery; shim already proven at 4 sites
Staff resistance to new workflow	Medium	Branch champions trained in phase 5; four weeks parallel running
Key personnel loss	Low	Contractual commitment to named architect and migration lead

6. Support Model, Experience and References

Support is 24x7 for priority 1 incidents with a one hour response target and a four hour resolution target, and business hours for priority 3 and 4. A named service delivery manager holds a monthly service review with the Service. Service credits apply against the annual support charge where targets are missed in two consecutive months.

Apex Systems was founded in 2009 and employs 240 staff. We have delivered library and archive systems to seven public sector clients. Two comparable references:

- Northgate Borough Libraries, 9 branches, 110,000 members, delivered 2023, migrated 14 years of circulation history with zero reconciliation variance
- Weston County Archive Service, 4 sites, delivered 2024, ISO 27001 aligned deployment with council SSO federation